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MCU SUB ASSEMBLY STATION

FOR

TATA MOTORS LTD.



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LATEST VERSION OF THIS MACHINE MANUAL

RELIABLE PERFORMANCE

BUILT FOR EXCELLENCE

SUPPORT YOU CAN TRUST

Your Growth Our Priority

PROWESS
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USER MANUAL

SPECIAL PURPOSE MACHINE (SPM)

MCU SUB ASSEMBLY STATION



CUSTOMER
Tata Motors Ltd.

MANUFACTURER
Prowess Autotech Pvt. Ltd.

DATE
10-10-2026

READ THIS MANUAL COMPLETELY BEFORE OPERATING THE MACHINE.

SAFE OPERATION | PRECISE PERFORMANCE | PREMIUM QUALITY | EASY MAINTENANCE | RELIABLE RESULTS

PROWESS AUTOTECH PVT. LTD.

LOCATION : RUDRAPUR, UTTARAKHAND
PIN CODE : 263153

+91 9917474110
prowessautotech@gmail.com

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3. OPERATING INSTRUCTIONS

This section describes the operating procedure for
MCU SUB ASSEMBLY STATION SPM.

NOTE
The machine cycle will proceed only when the part is correctly positioned, both push buttons are pressed together, and torque confirmation is accepted on the HMI.

3.1 START-UP PROCEDURE

1. Switch ON the main power supply.
2. Ensure the compressed air supply is ON and pressure is within the approved operating range.
3. Check that the fixture and resting pads are clean and free of obstruction.
4. Confirm the correct component is available for loading.
5. Check that the torque wrench is ready and set to the specified torque.
6. Switch ON the control panel and check the HMI for alarms or faults.
7. Ensure the fixture is in the unclamped/ home condition before loading.

3.2 OPERATING CYCLE

1. Load the component onto the fixture resting pads.
2. Position the component correctly.
3. Press both cycle start push buttons simultaneously to clamp.
4. Tighten the bolts using the torque wrench.
5. Verify torque confirmation on the HMI.
6. After accepted confirmation, the fixture automatically de-clamps
7. Unload the completed part and load the next component.

3.3 OPERATOR GUIDELINES

- Keep hands clear of clamping points and moving parts.
- Press both push buttons together; never bypass the safety function.
- Do not adjust or reposition the component while clamped.
- Use the torque wrench and specified torque value only.
- Confirm torque status on the HMI before unloading.
- Do not force the fixture open or interfere with automatic de-clamping.
- Report abnormal noise, air leakage, or HMI faults.
- Only authorized personnel may perform maintenance.

3.4 EMERGENCY STOP PROCEDURE

1. Press EMERGENCY STOP immediately if an unsafe condition occurs.
2. Keep clear and inform the supervisor/maintenance team.
3. Resolve the fault and reset Emergency Stop only when safe.
4. Restart the cycle only after checks and authorization.

IMPORTANT
Never bypass safety devices. Keep hands clear of the fixture during clamping and de-clamping.
Follow site lockout/tagout rules before maintenance.

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6. MAINTENANCE SCHEDULE

Regular maintenance ensures smooth operation, longer machine life and consistent performance of the MCU SUB ASSEMBLY STATION.

NOTE

- Follow the maintenance intervals mentioned below.
- Use recommended tools and genuine spares where required.
- If any abnormal noise, vibration or fault is observed, stop the machine and inform the supervisor immediately.

INTERVAL	TASK / CHECK POINT	DETAILS	RESPONSIBLE
DAILY (Before Start)	• Clean machine surface. • Check air pressure. • Check push buttons (both hands). • Check HMI display. • Visual check of fixture & resting pads. • Check for air leaks / unusual noise.	Remove dust, chips and debris. Ensure 5 – 6 Bar pressure. Both buttons should work together and not stick. HMI should power ON and display normally. Pads should be clean and properly seated. Listen for abnormal sound / air leak.	Operator
WEEKLY (Every 7 Days)	• Lubricate moving parts (guide rails, etc.). • Check air filter / regulator. • Check pneumatic cylinder operation. • Inspect cables & connectors. • Check torque wrench calibration (visual / functional).	Use recommended lubricant (light machine oil). Clean filter bowl and drain water. Smooth operation, no air leakage. Ensure wiring is tight and undamaged. Verify wrench works properly and gives correct torque (as per set value).	Maintenance (Mechanical + Electrical)
MONTHLY	• Check solenoid valves (operation). • Inspect hoses and fittings. • Check fixture alignment & clamping pads. • Tighten loose bolts and fasteners. • Check electrical panel interior (dust).	Ensure proper response and no leakage. No cracks / wear, proper routing. Fixture should sit correctly, pads undamaged. Check all hardware is tight. Remove dust (if required).	Maintenance (Mechanical + Electrical)
3 MONTHS	• Replace air filter element (if required). • Check pneumatic seals / cylinder rod. • Inspect wiring, relays and contactors. • Verify HMI touch response.	Use genuine filter element. Check for wear, leakage or damage. Clean and tighten terminals. Check display, touch and buttons.	Maintenance (Mechanical + Electrical)
6 MONTHS	• Check complete pneumatic system (pressure, leaks, hoses). • Inspect clamping cylinders for wear. • Check push button safety function. • Check torque wrench accuracy.	Ensure stable pressure and no leakage. Look for seal wear or smooth movement. Both buttons and safety circuit must work properly. Calibrate / verify torque value (if required).	Maintenance (Mechanical + Electrical)
YEARLY	• Full machine inspection & cleaning. • Replace worn-out parts (as required). • Check electrical panel components. • Verify safety interlocks and emergency stop.	Remove dust and tighten all fittings. Replace damaged parts (pads, seals, switches, etc.). Check wiring, terminals, and components. Test E-Stop and safety function.	Maintenance (Authorized / Qualified)

USE GENUINE SPARES
For better performance and longer life.

KEEP IT CLEAN
Dust and chips reduce accuracy and life.

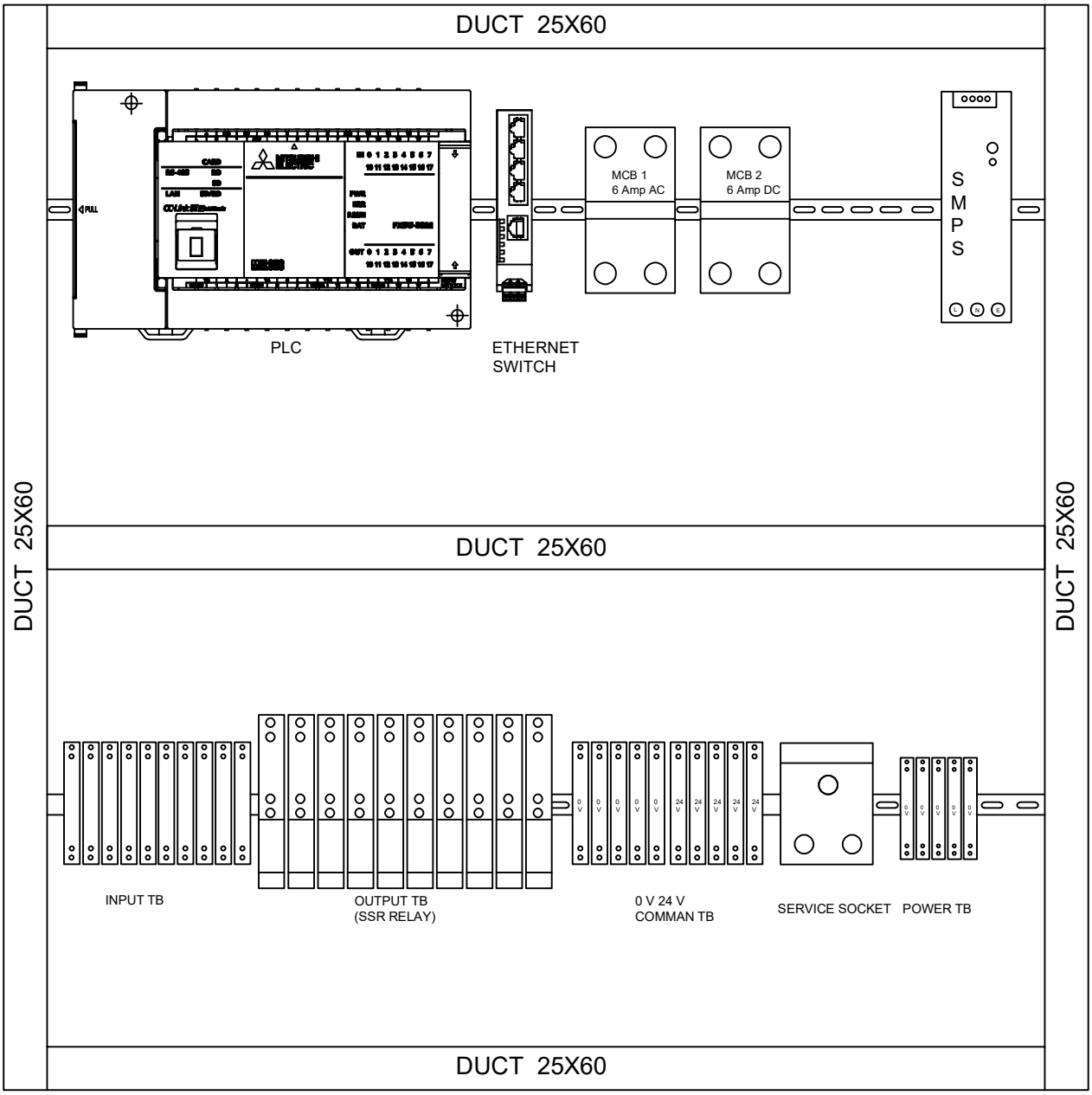
DO NOT BYPASS SAFETY INTERLOCKS
They are for your safety.

FOLLOW TORQUE VALUES
For reliable clamping and quality output.

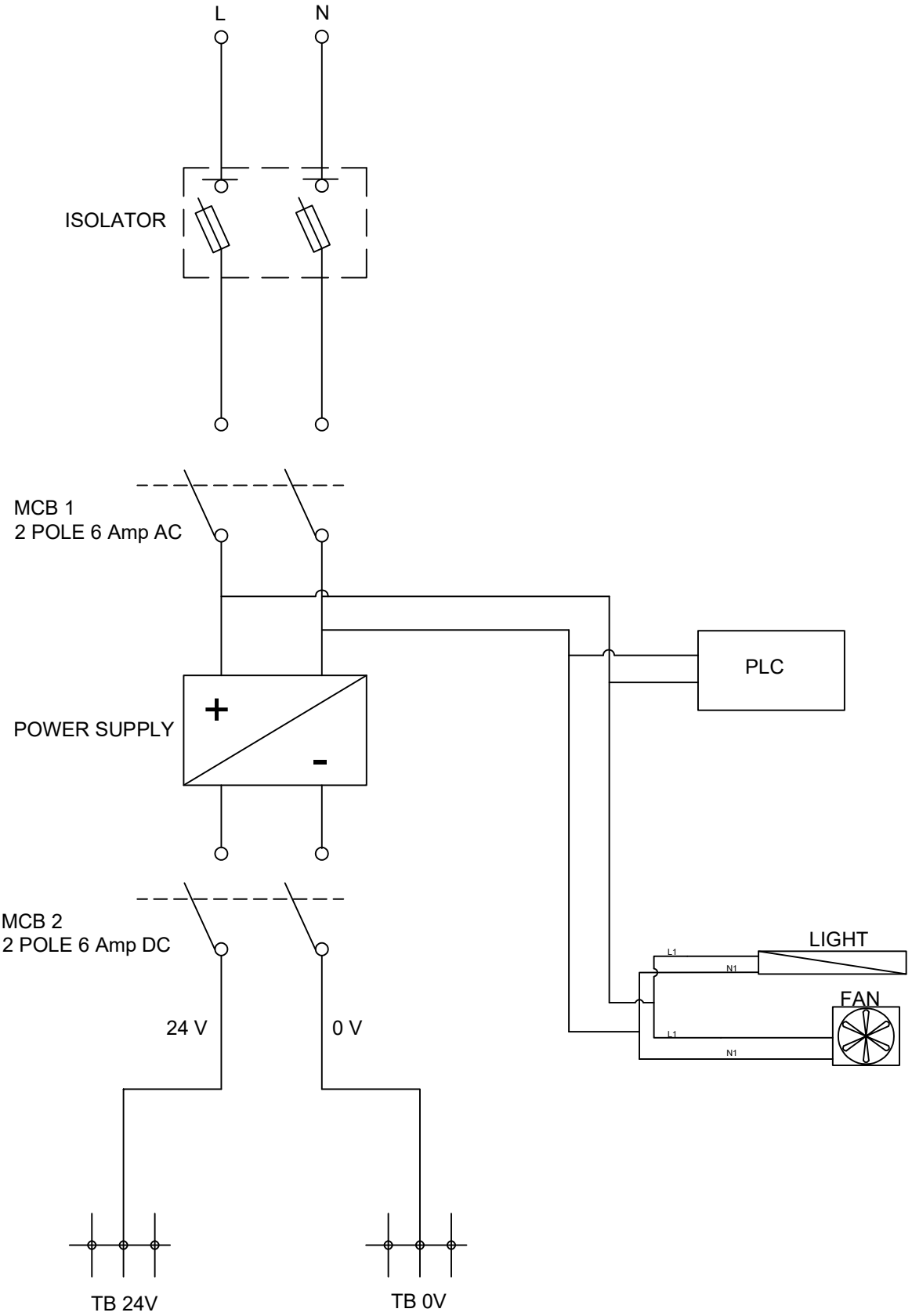
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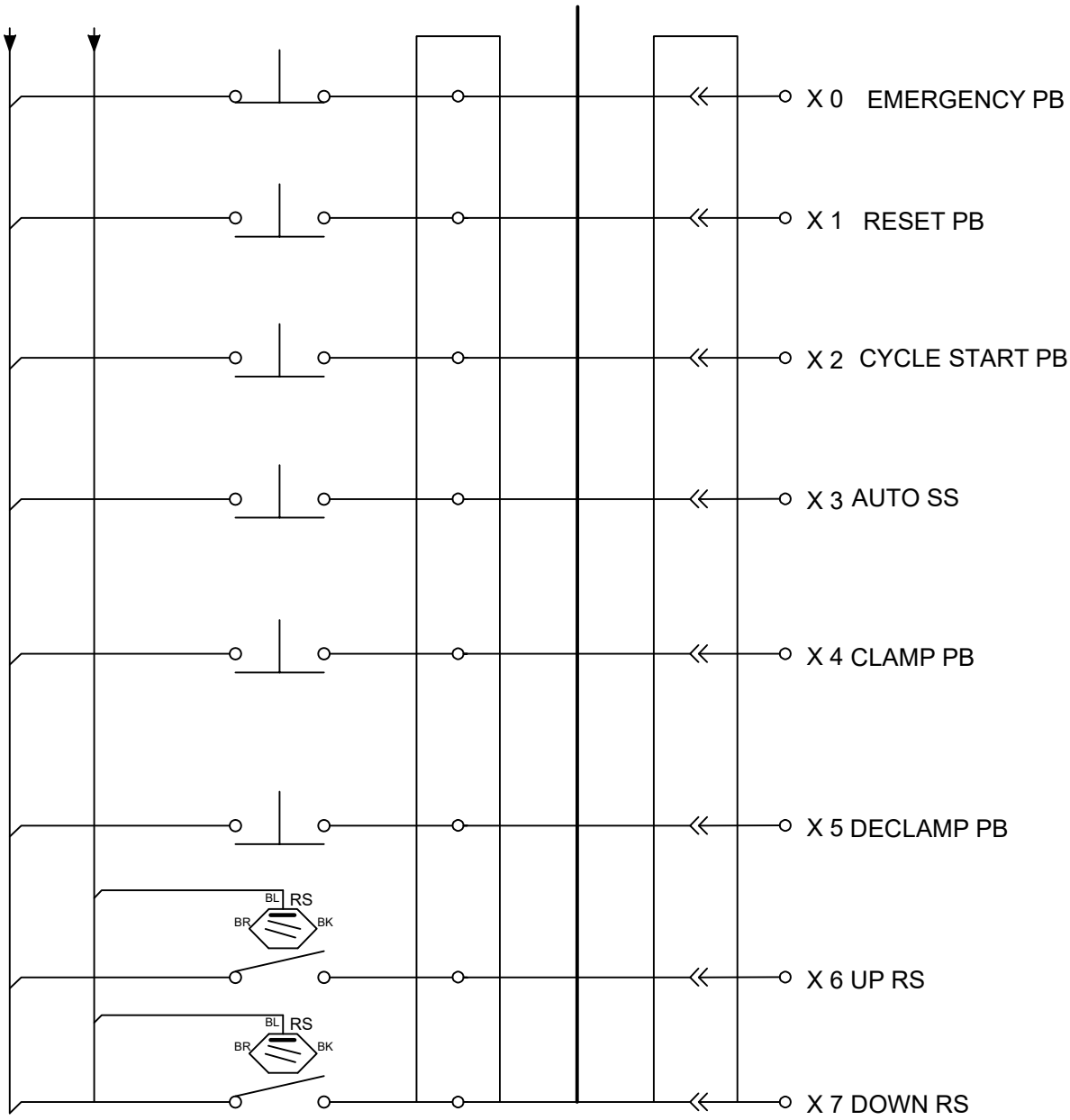


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Area		Approved	Mr. Omprakash	MCU SUB ASSEM STATION	
Date		Date	10-10-2026		
Name		M/C Type	MAIN PANEL		Sheet No: 05



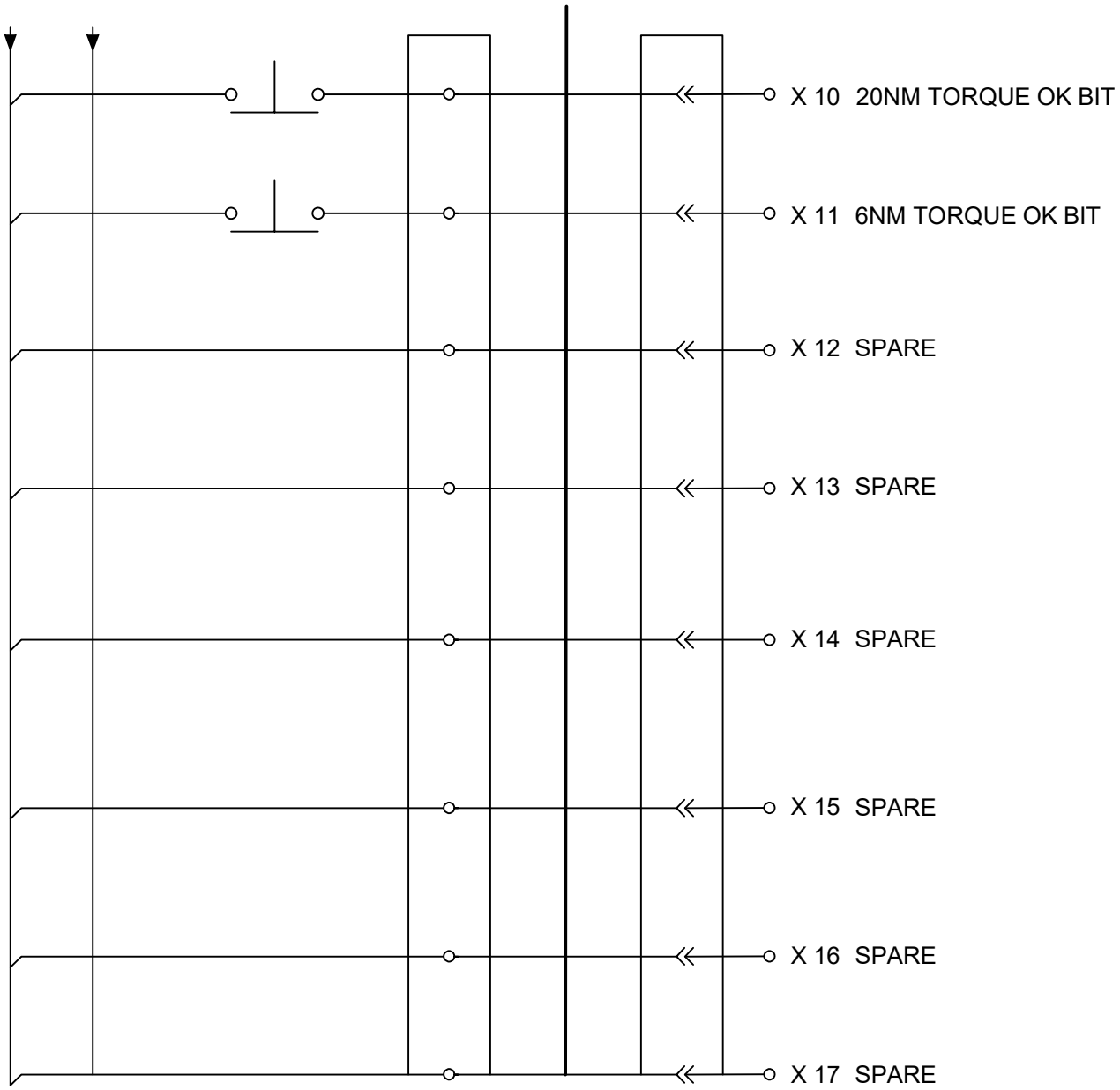
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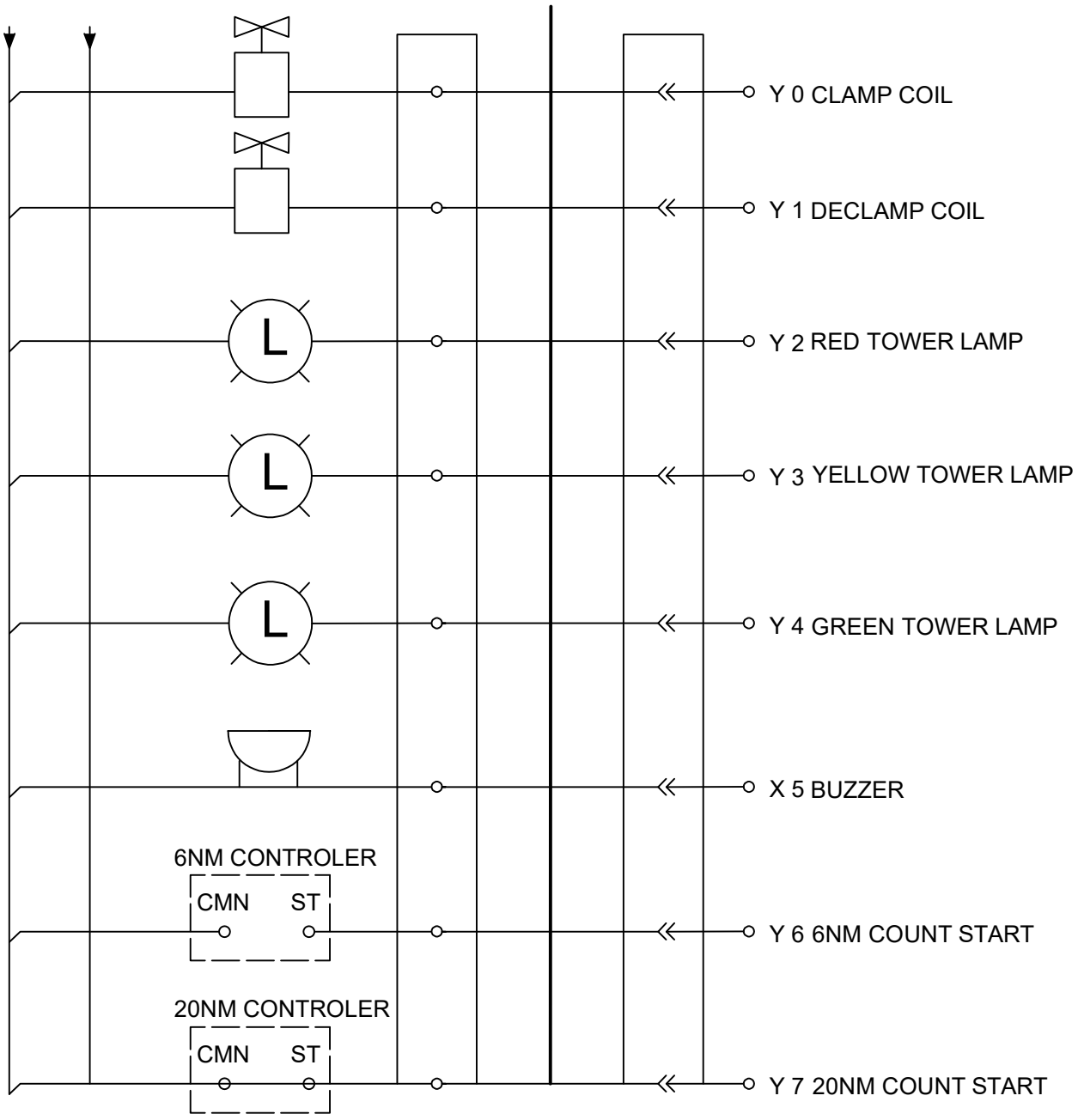
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Area		Approved	Mr. Omprakash	MCU SUB ASSEM STATION	
Date		Date	10-10-2026		
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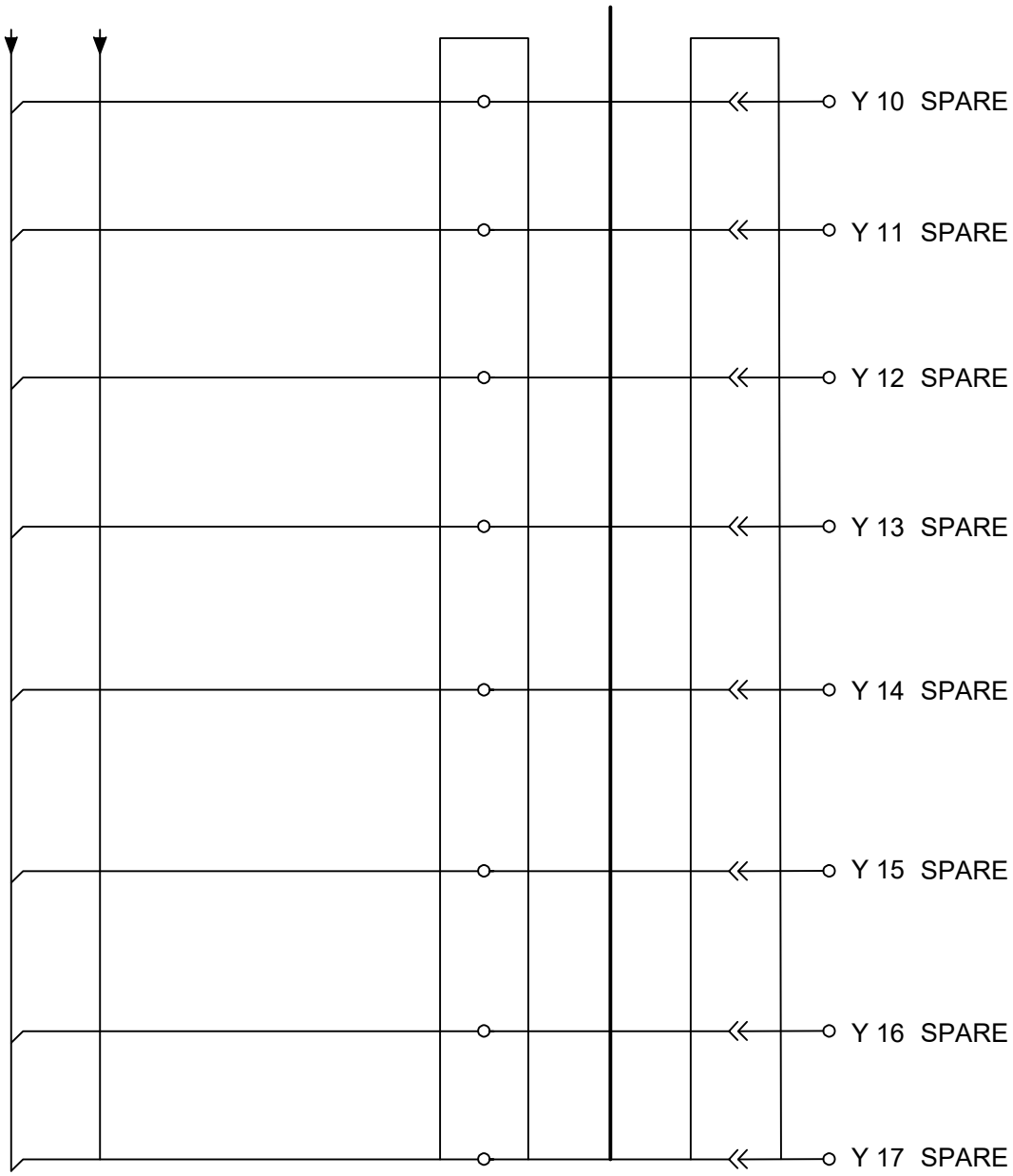
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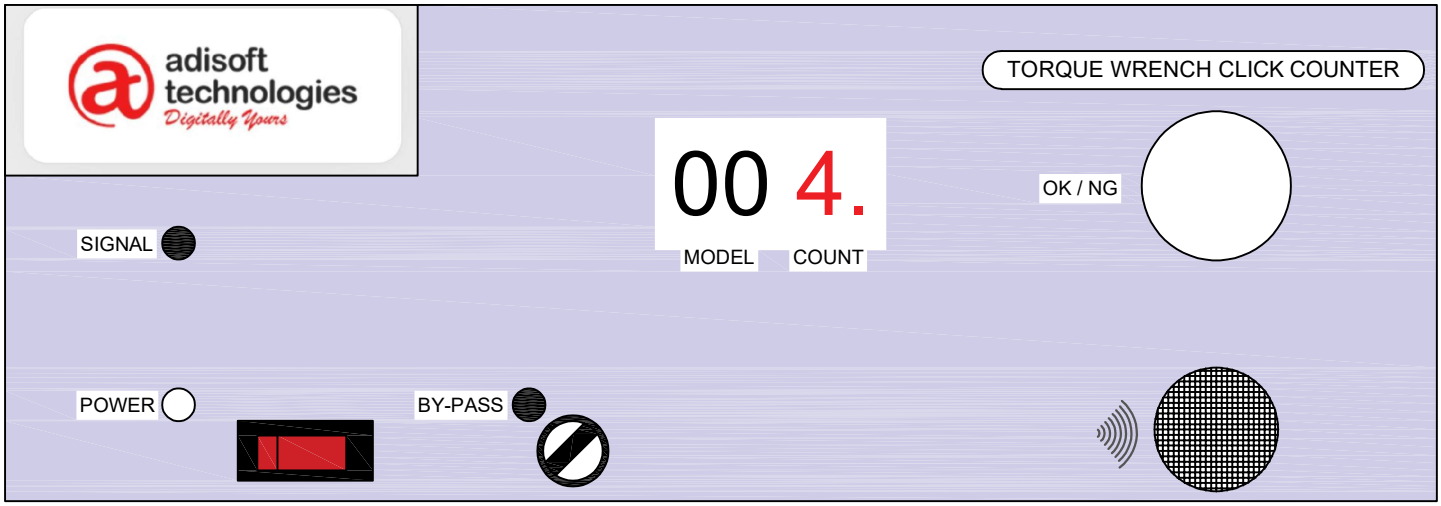


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Date		Date	10-10-2026		
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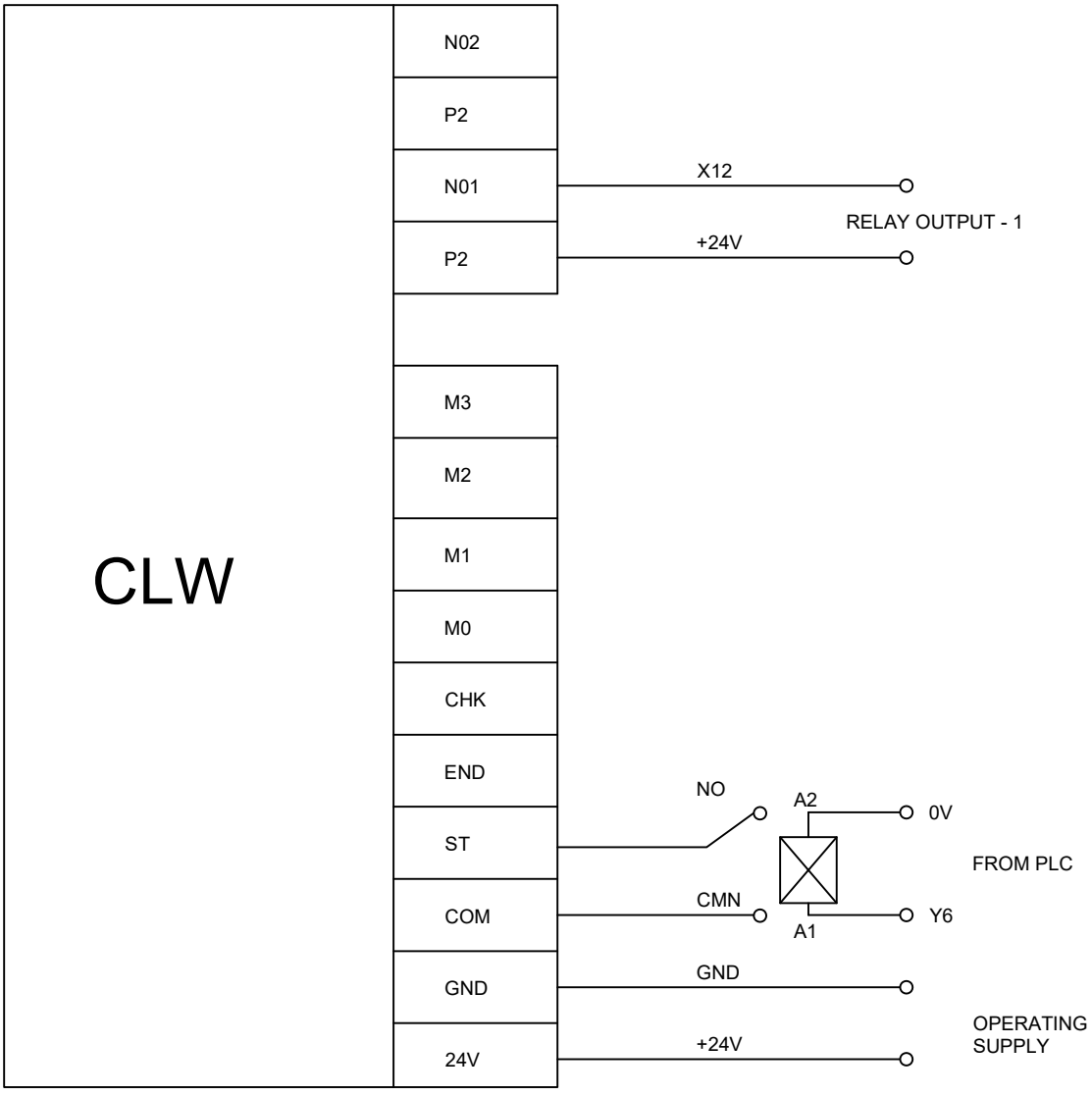
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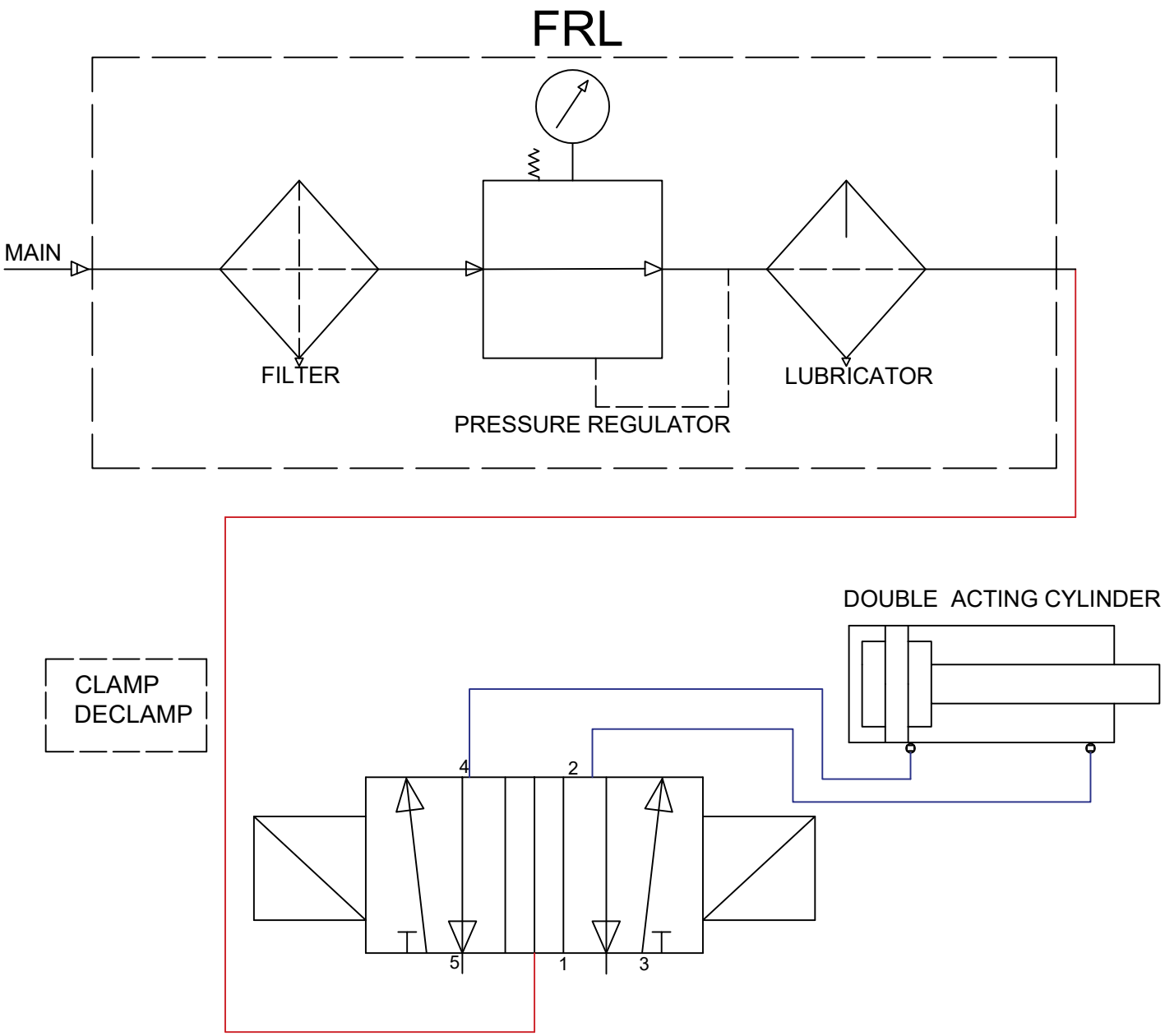
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Area		Approved	Mr. Omprakash	MCU SUB ASSEM STATION	
Date		Date	10-10-2026		
Name		M/C Type	OUTPUT LIST		Sheet No: 04



EXTERNAL CONNECTION



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Area		Approved	Mr. Omprakash	MCU SUB ASSEM STATION	
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Name		M/C Type	TORQUE WRENCH CLICK CTR		Sheet No: 07

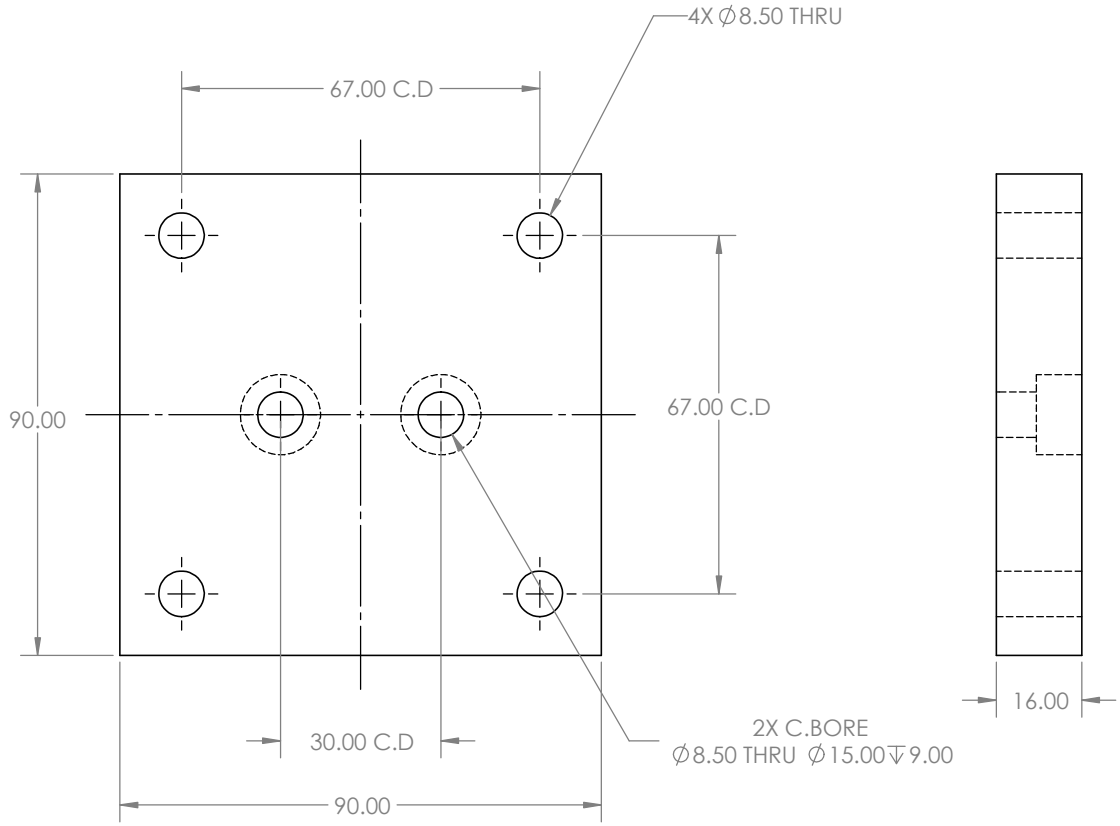


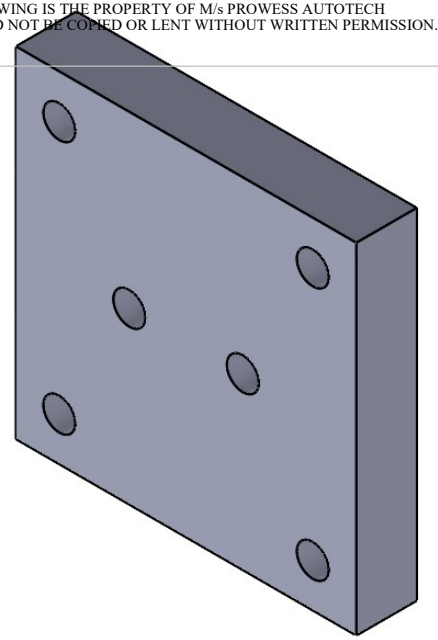
5/2. Directional control valve

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Area		Approved	Mr. Omprakash	MCU SUB ASSEM STATION	
Date		Date	10-10-2026		
Name		M/C Type	PNEUMATIC		Sheet No: 05

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Checked	Chanchal Pandey		
Approved	Om Prakash		Appr. Date
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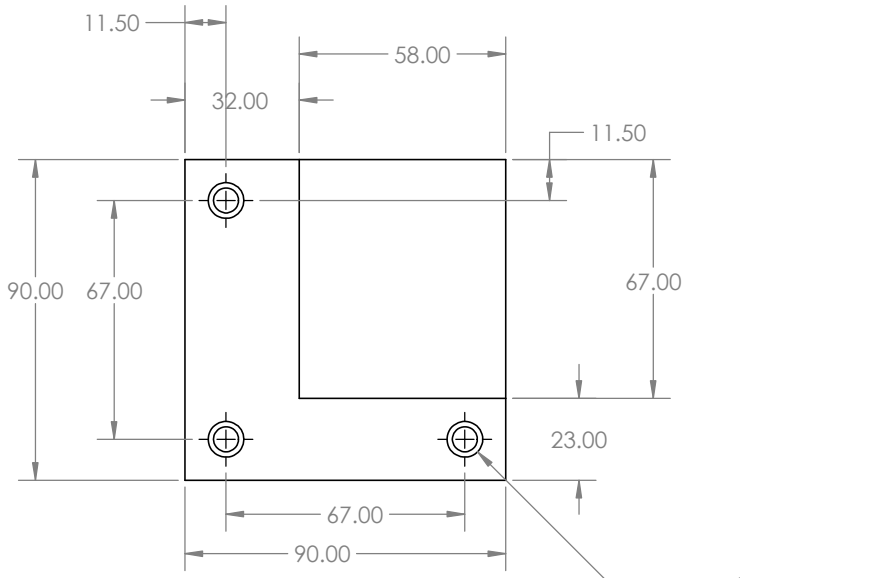
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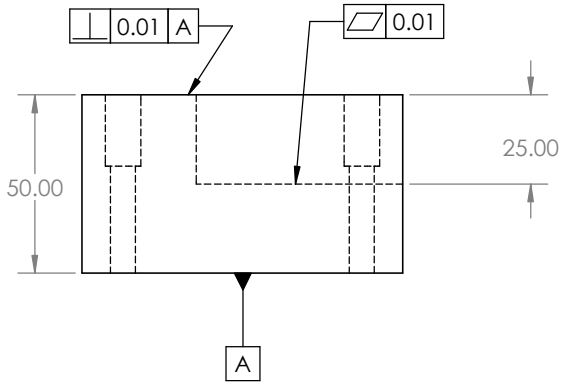
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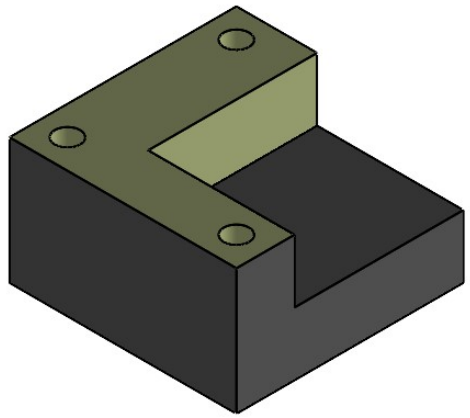
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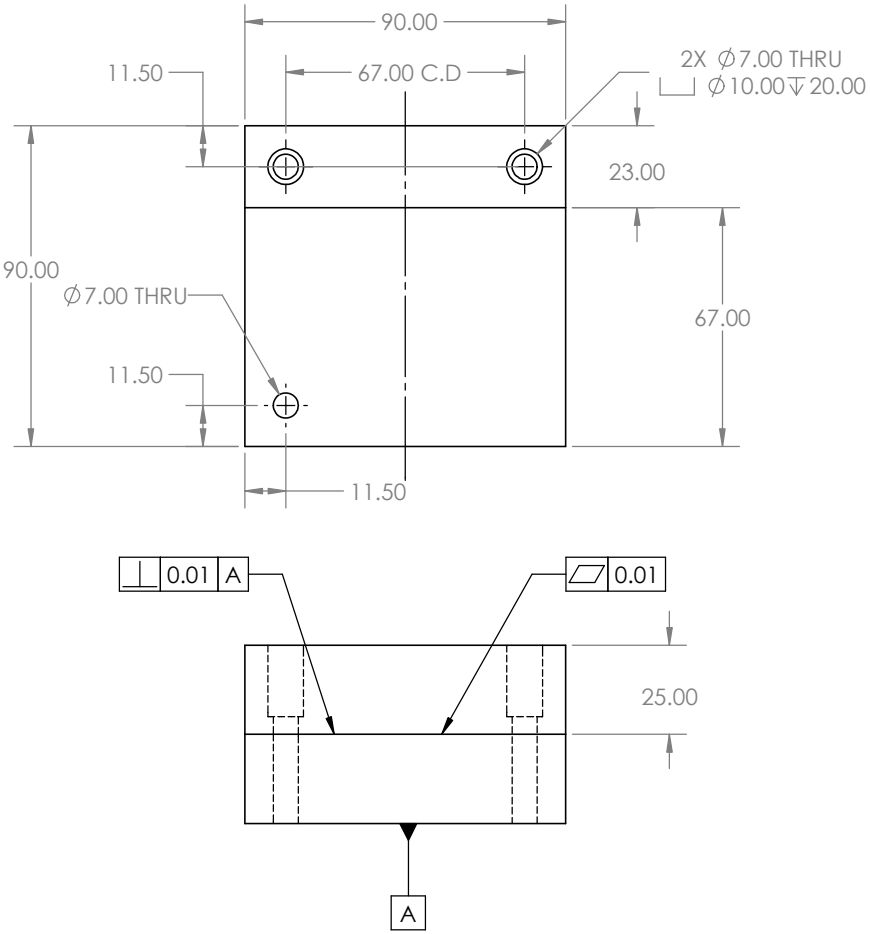
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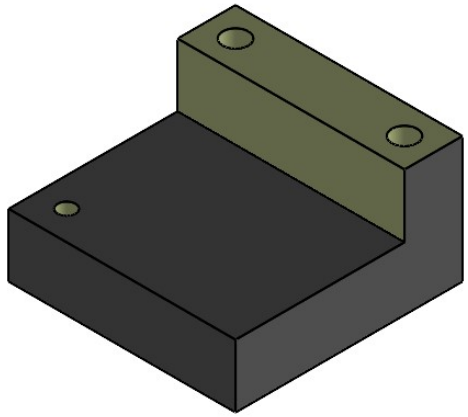
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Approved	Om Prakash		Appr. Date
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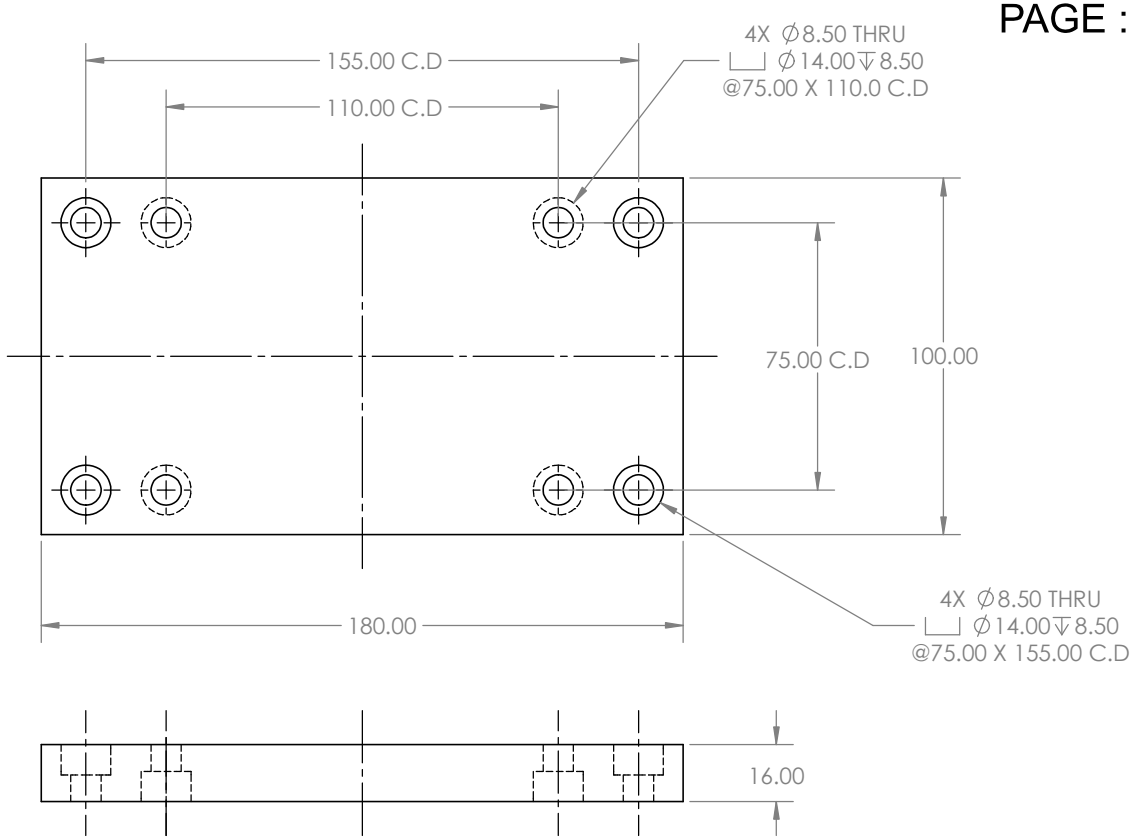
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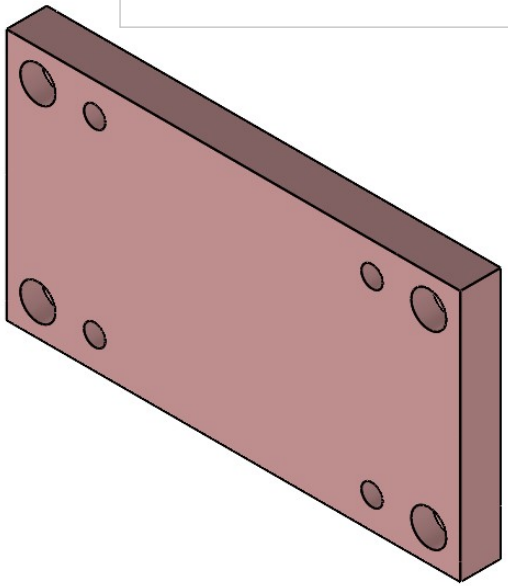
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Approved	Om Prakash		Appr. Date
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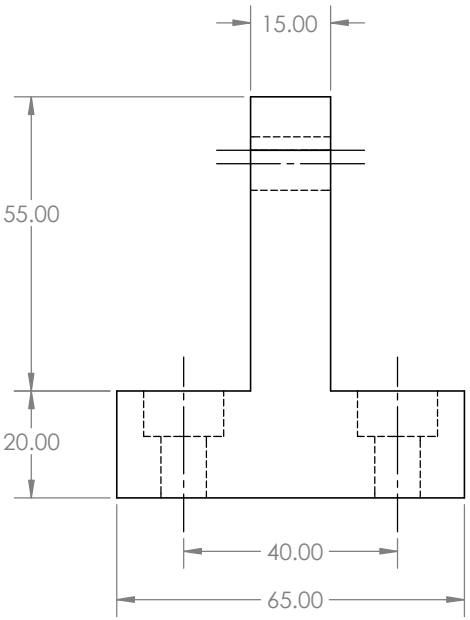
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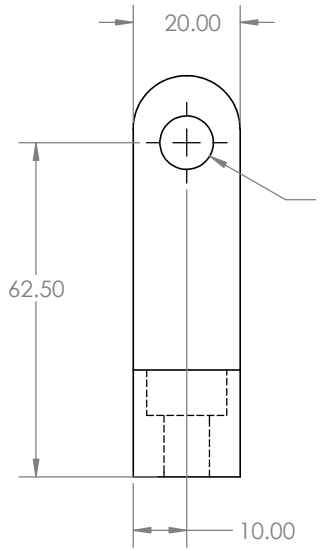
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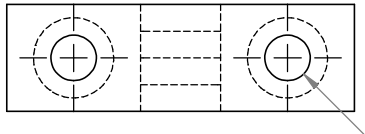
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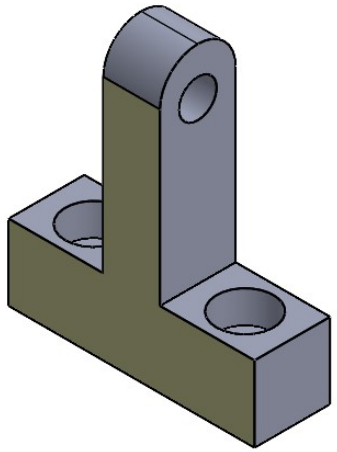
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Approved	Om Prakash		Appr. Date
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TITLE
MCU SUB ASSEM STATION

Ref Drawing No.

Drawing No.

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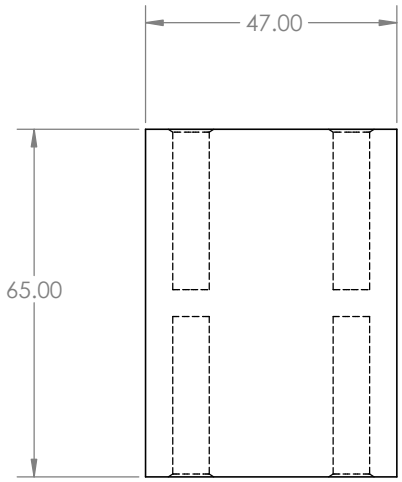
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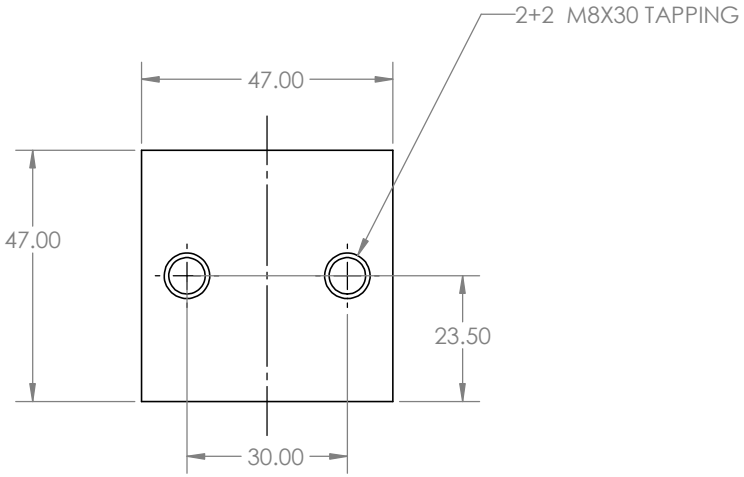
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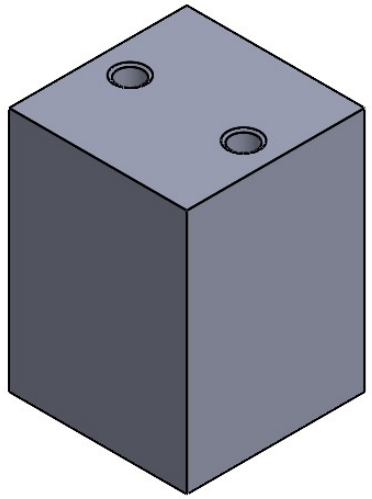
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MATERIAL = BRIGHT BAR
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Rev						
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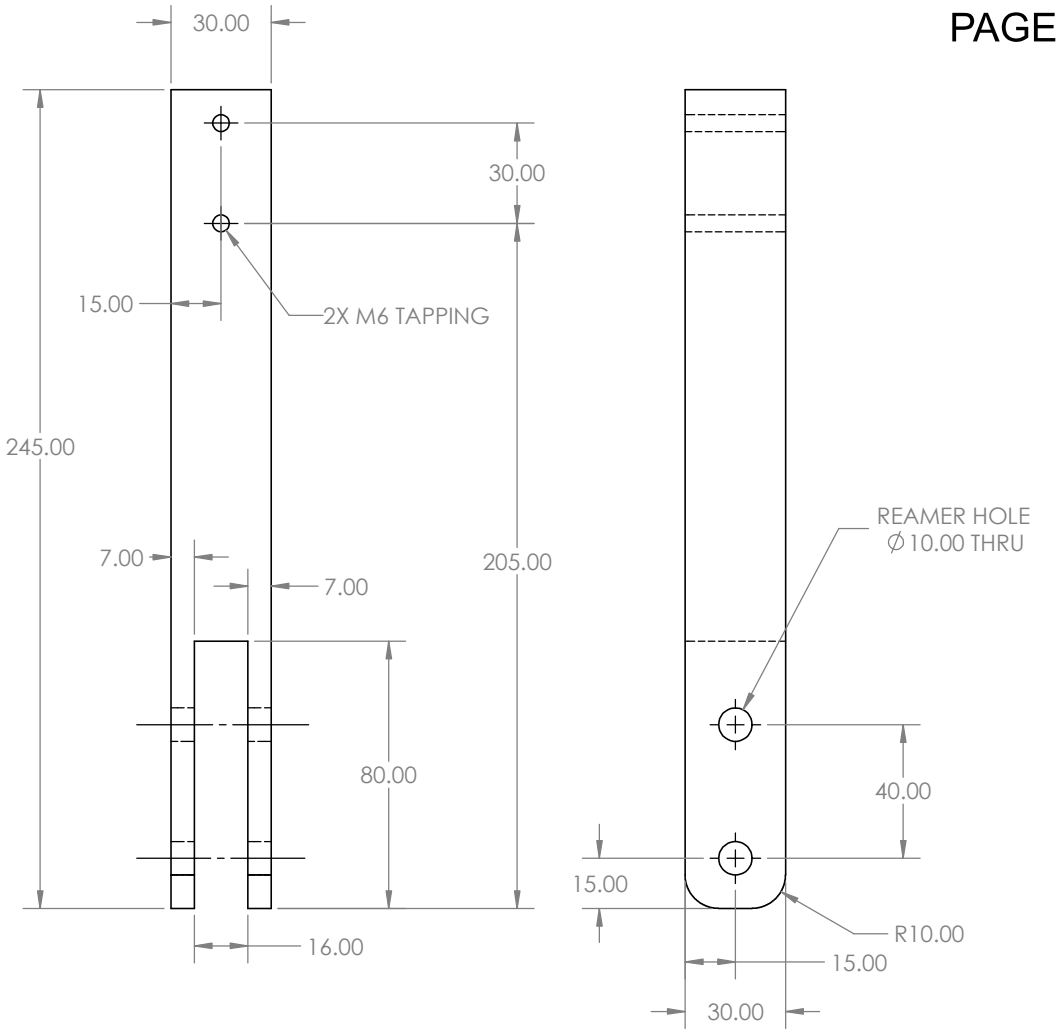
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GENERAL GEOMETRIC TOLERANCE
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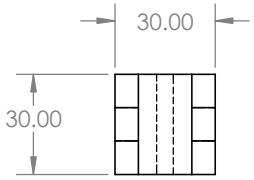
Key Characteristics | S Safety Characteristics | Critical Characteristics | Important Characteristics

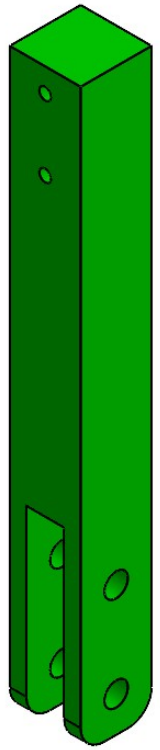
PROWESS AUTOTECH PVT. LTD. OUTSOURCE YOUR PAIN

DO NOT SCALE THE DRG.
ASK IF IN DOUBT.

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MATERIAL = BRIGHT BAR
QTY = 1 NOS

Scale	Drawing No.	Description	Case Depth	Weight	Qty.	Assy Drg. No.
1:1						

Finish		Sign.	
Drawn	Christopher		Projection
Checked	Chanchal Pandey		
Approved	Om Prakash		Appr. Date
TYPE		Material	

TITLE
MCU SUB ATATION ASSEM

Ref Drawing No.

Drawing No.

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MEASURE WITHOUT TOLERANCE INDICATION :
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GENERAL GEOMETRIC TOLERANCE
REFERENCE ISO 2768-2 TOLERANCE CLASS: L

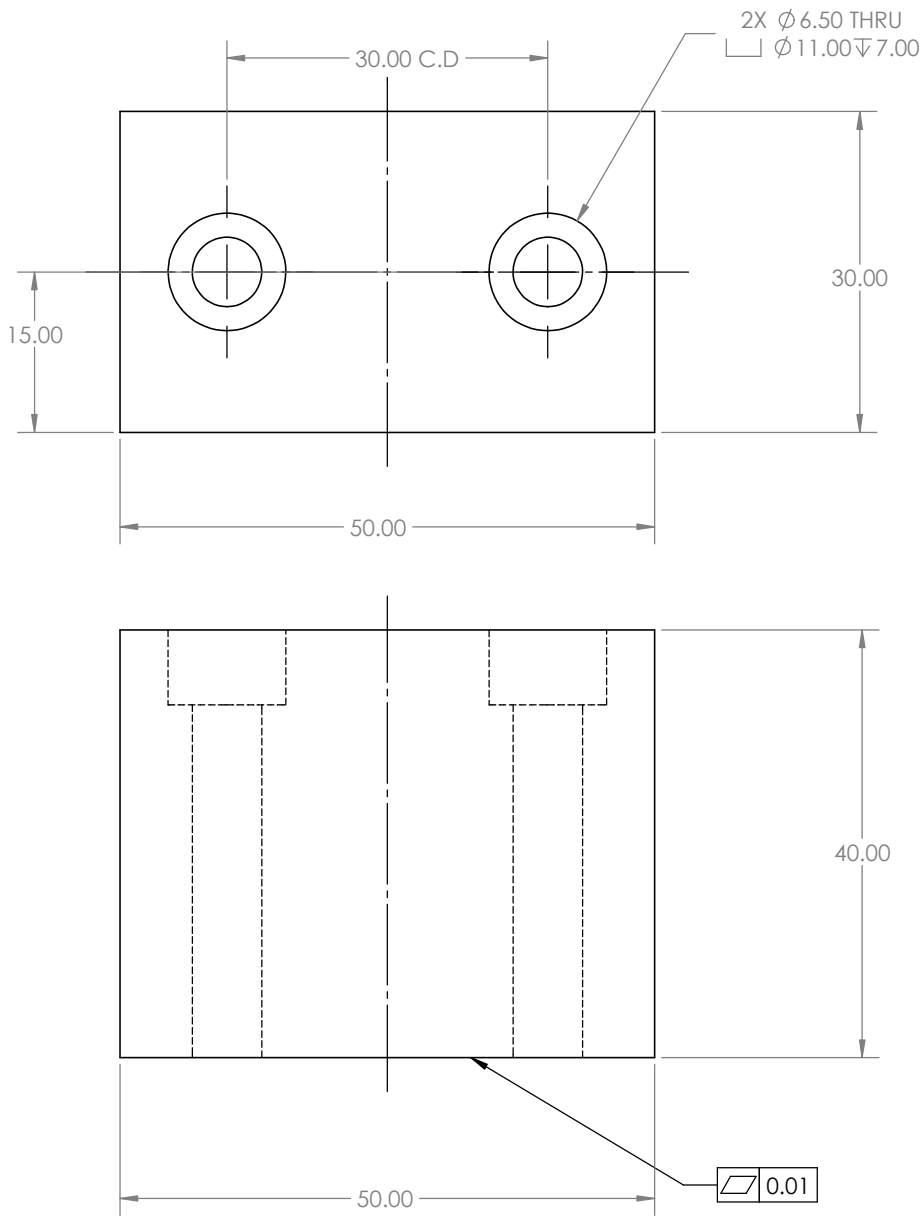
Key Characteristics | S Safety Characteristics | Critical Characteristics | Important Characteristics

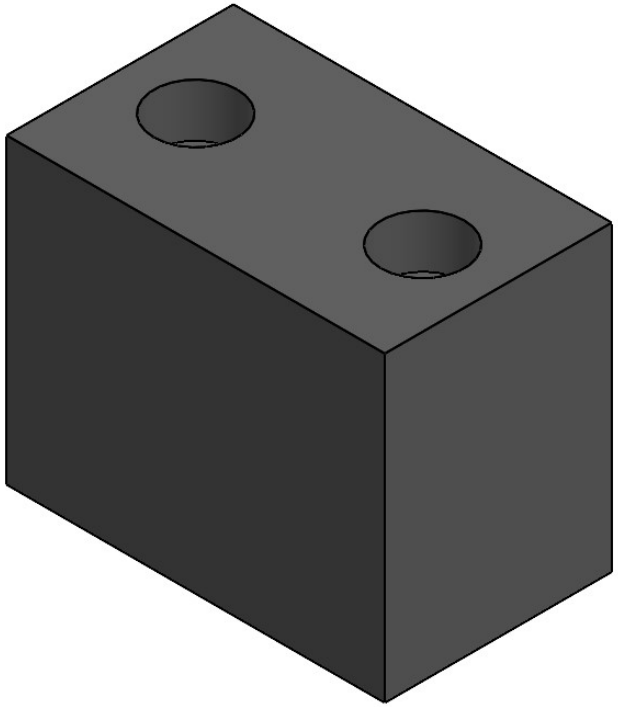
PROWESS AUTOTECH PVT. LTD. OUTSOURCE YOUR PAIN

Rev

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MATERIAL = TEFFLON
QTY = 1 NOS

Rev						
Scale	Drawing No.	Description	Case Depth	Weight	Qty.	Assy Drg. No.
1:1						

Finish		Sign.	
Drawn	Christopher		Projection
Checked	Chanchal Pandey		
Approved	Om Prakash		Appr. Date
TYPE		Material	

Prowess Autotech Pvt. Ltd.

MEASURE WITHOUT TOLERANCE INDICATION :
REFERENCE ISO 2768-1 TOLERANCE CLASS:M
GENERAL GEOMETRIC TOLERANCE
REFERENCE ISO 2768-2 TOLERANCE CLASS: L

TITLE
MCU SUB ASSEM STATION

Ref Drawing No.

Drawing No.

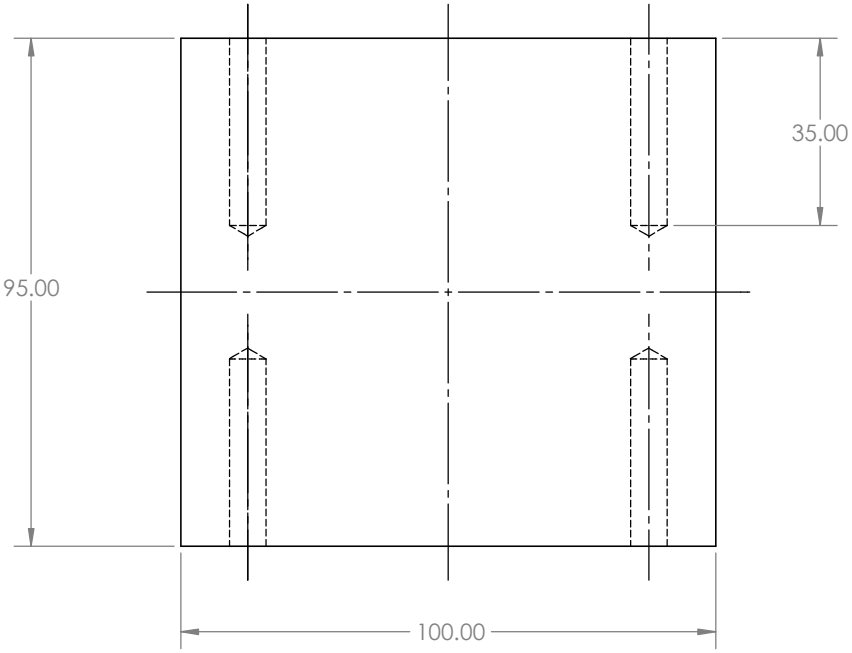
Key Characteristics (S) Safety Characteristics Critical Characteristics Important Characteristics

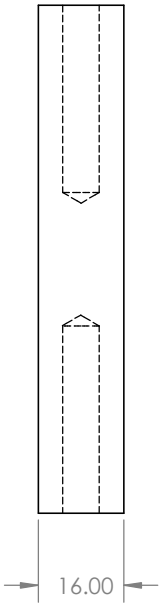
PROWESS AUTOTECH PVT. LTD. OUTSOURCE YOUR PAIN

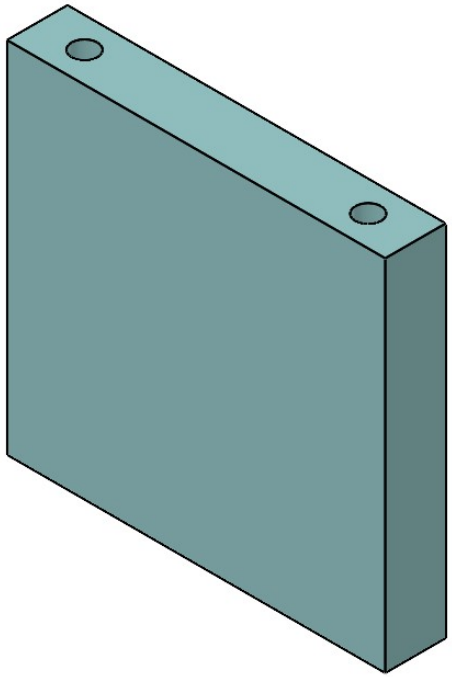
DO NOT SCALE THE DRG.
ASK IF IN DOUBT.

PAGE : 20

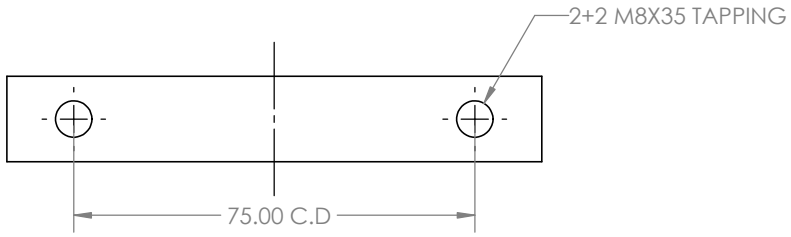
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MATERIAL = MS
QTY = 2 NOS



Rev						
Scale	Drawing No.	Description	Case Depth	Weight	Qty.	Assy Drg. No.
1:1						

Finish		Sign.	
Drawn	Christopher		Projection
Checked	Chanchal Pandey		
Approved	Om Prakash		Appr. Date
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Prowess Autotech Pvt. Ltd.

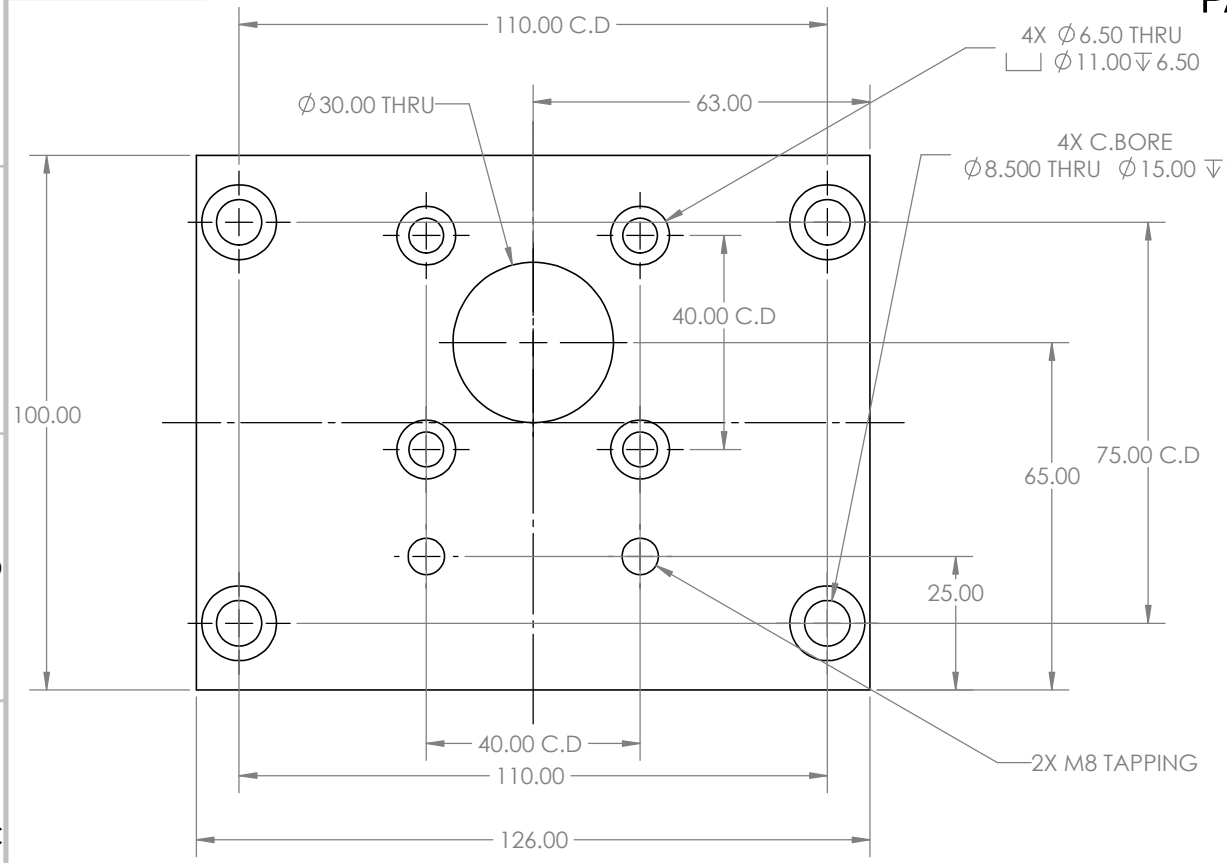
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REFERENCE ISO 2768-2 TOLERANCE CLASS: L

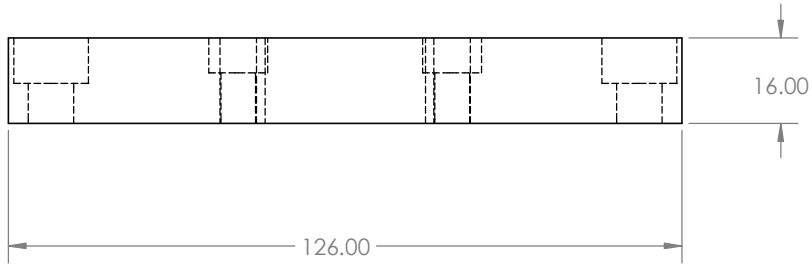
Key Characteristics (S) Safety Characteristics Critical Characteristics Important Characteristics

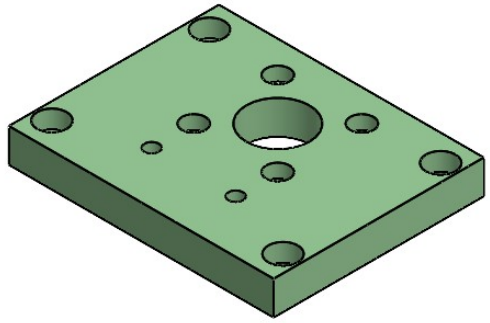
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MATERIAL = MS
QTY = 1NOS

Scale	Drawing No.	Description	Case Depth	Weight	Qty.	Assy Drg. No.
1:1						

Rev

Finish		Sign.	
Drawn	Christopher		Projection
Checked	Chanchal Pandey		
Approved	Om Prakash		Appr. Date
TYPE		Material	

Prowess Autotech Pvt. Ltd.

MEASURE WITHOUT TOLERANCE INDICATION :
REFERENCE ISO 2768-1 TOLERANCE CLASS:M
GENERAL GEOMETRIC TOLERANCE
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TITLE
MCU SUB ASSEM STATION

Ref Drawing No.

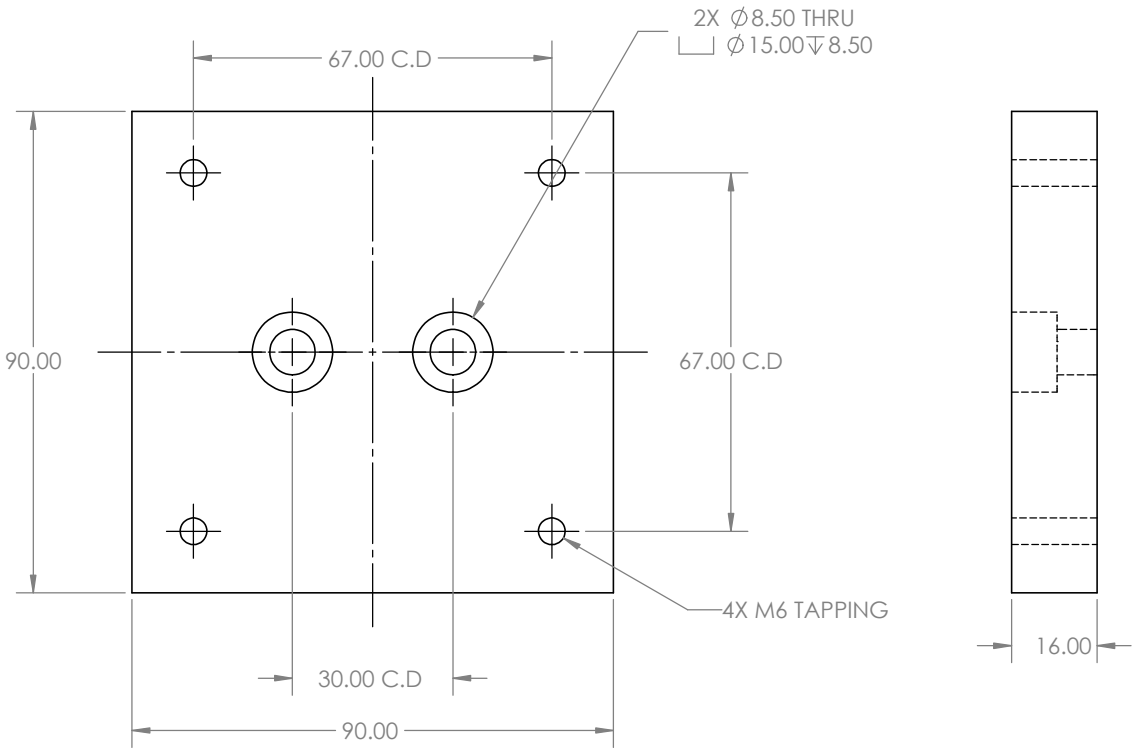
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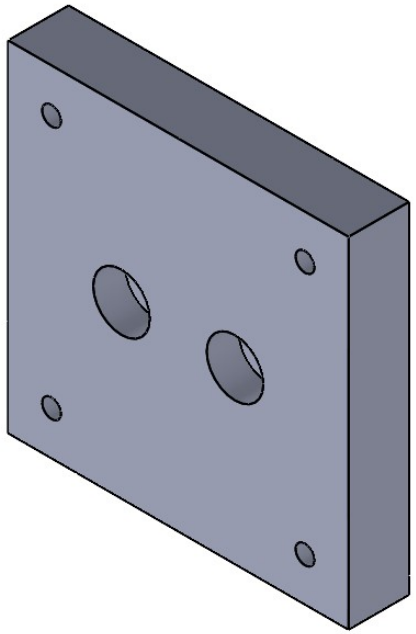
Key Characteristics | S Safety Characteristics | Critical Characteristics | Important Characteristics

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MATERIAL = MS
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Rev						
Scale	Drawing No.	Description	Case Depth	Weight	Qty.	Assy Drg. No.
1:1						

Finish		Sign.	
Drawn	Christopher		Projection
Checked	Chanchal Pandey		
Approved	Om Prakash		Appr. Date
TYPE		Material	
TITLE MCU SUB STATION ASSEM		Ref Drawing No.	
		Drawing No.	

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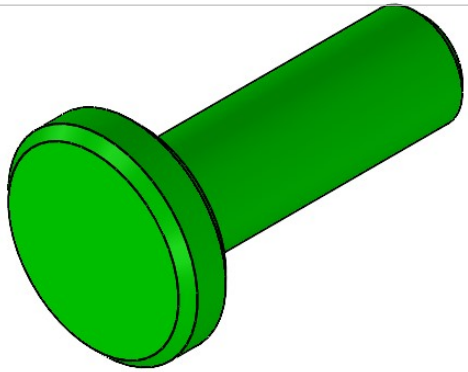
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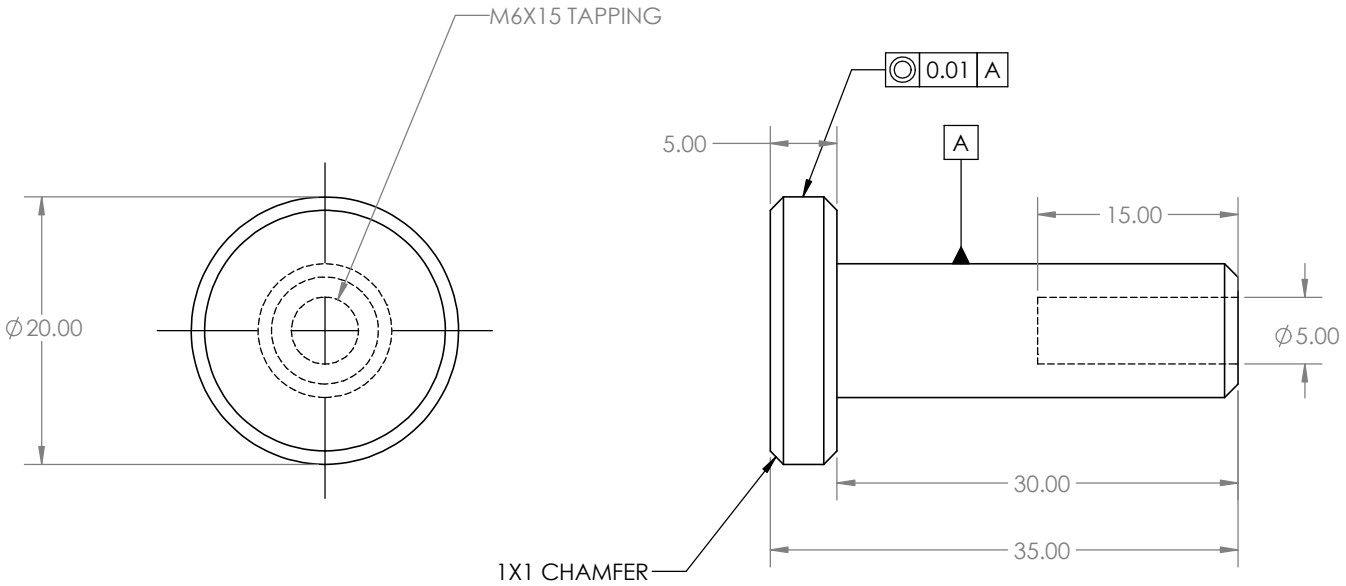
Key Characteristics (S) Safety Characteristics Critical Characteristics Important Characteristics

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MATERIAL = EN8
QTY = 2 NOS

Rev						
Scale	Drawing No.	Description	Case Depth	Weight	Qty.	Assy Drg. No.
1:1						

Finish		Sign.	
Drawn	Christopher		Projection
Checked	Chanchal Pandey		
Approved	Om Prakash		Appr. Date
TYPE		Material	
TITLE MCU SUB ASSEMSTATION		Ref Drawing No.	
		Drawing No.	

Prowess Autotech Pvt. Ltd.

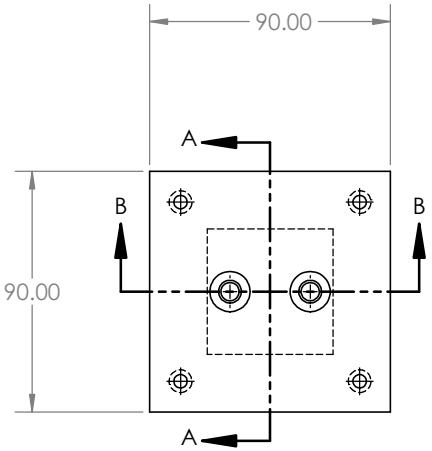
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REFERENCE ISO 2768-2 TOLERANCE CLASS: L

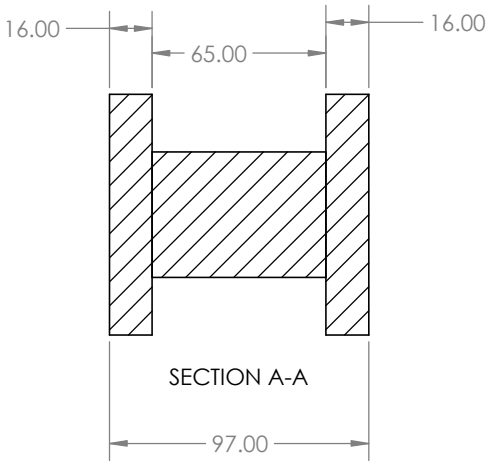
Key Characteristics | S Safety Characteristics | Critical Characteristics | Important Characteristics

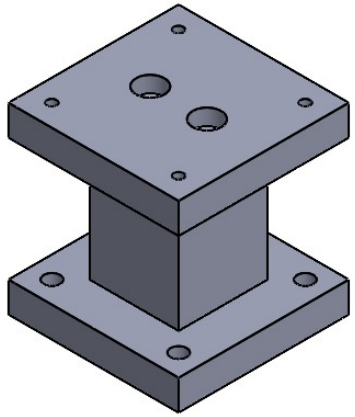
PROWESS AUTOTECH PVT. LTD. OUTSOURCE YOUR PAIN

DO NOT SCALE THE DRG.
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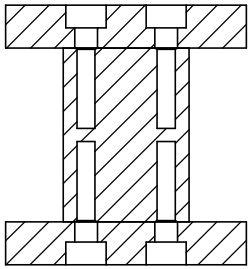
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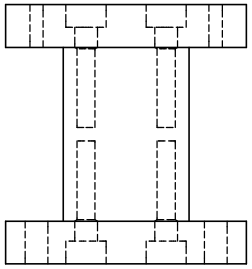




TEFFLON MTG ASSEMBLY VIEW
QTY 4 NOS



SECTION B-B



Rev						
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Finish		Sign.	
Drawn	Christopher		Projection
Checked	Chanchal Pandey		
Approved	Om Prakash		Appr. Date
TYPE		Material	
TITLE PART NAME			Ref Drawing No.
			Drawing No.

Prowess Autotech Pvt. Ltd.

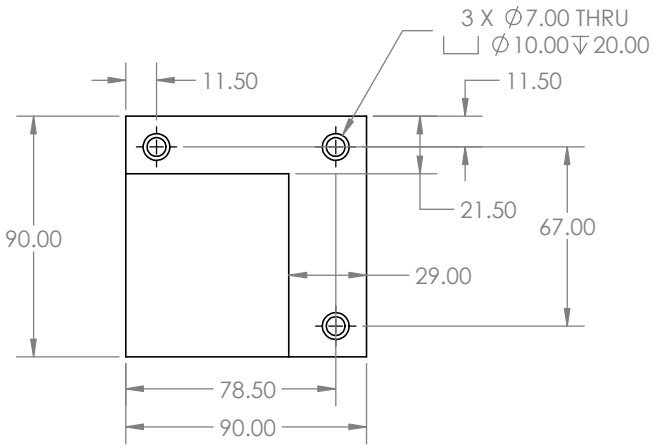
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REFERENCE ISO 2768-2 TOLERANCE CLASS: L

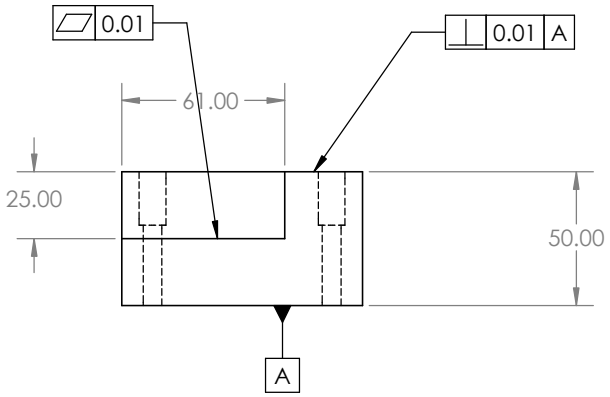
Key Characteristics | S Safety Characteristics | Critical Characteristics | Important Characteristics

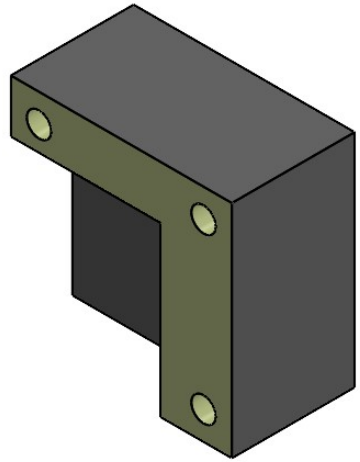
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MATERIAL = TEFFLON
QTY = 2 NOS

Scale	Drawing No.	Description	Case Depth	Weight	Qty.	Assy Drg. No.
1:1						

Rev

Finish		Sign.	
Drawn	Christopher		Projection
Checked	Chanchal Pandey		
Approved	Om Prakash		Appr. Date
TYPE		Material	

Prowess Autotech Pvt. Ltd.

MEASURE WITHOUT TOLERANCE INDICATION :
REFERENCE ISO 2768-1 TOLERANCE CLASS:M
GENERAL GEOMETRIC TOLERANCE
REFERENCE ISO 2768-2 TOLERANCE CLASS: L

TITLE
MCU SUB ASSEMBLY STATION

Ref Drawing No.

Drawing No.

Key Characteristics (S) Safety Characteristics Critical Characteristics Important Characteristics

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PROWESS AUTOTECH PVT. LTD.

ELECTRICAL & MECHANICAL BOM

MCU SUB ASSEMBLY STATION | TATA MOTORS LTD.

SR. NO	NAME	PART NO	MAKE	QTY
1	PLC	FX5U-32MR	MITSUBISHI	1
2	HMI	GS2107-WTBD-N	MITSUBISHI	1
3	POWER SUPPLY	PS-EE-24G/24AC/24DC	PHOENIX	1
4	EMERGENCY SWITCH	—	OTHER	3
5	ETHERNET SWITCH	FL 1005N	PHOENIX	1
6	ISOLATOR SWITCH	LB116	SALZER	1
7	MCB 6Amp 2 pole DC	C60H-DC C 6A	SCHNEIDER	1
8	MCB 6Amp 2 pole AC	A9N2P06CGN	SCHNEIDER	1
9	DOUBLE TB	KUDD4	ELMEX	20
10	TERMINAL BLOCK	CTS2.5UN	CONNECT WELL	5
11	FAN 220V	—	REXNORD	1
12	LED 5W	CR-LTLS-230-A-6W	CRAFTECH	1
13	2 POS. SELECTOR SWITCH	—	OTHER	1
14	YELLOW PB	—	TEKNIC	1
15	GREEN PB	—	TEKNIC	1
16	RED PB	—	TEKNIC	10
17	NC BLOCK	S2	TEKNIC	3
18	NO BLOCK	S1	TEKNIC	3
19	TOWER LIGHT	LSL0301RAGAB	LEVZE ELECTRONIC	1
20	READ SWITCH	D-M9P2	SMC	4
21	FILTER	FPAV01-130*130	FTE	1
22	HOOTER LED	—	—	1
23	RELAY TB	—	PHOENIX CONTACT	8
24	1 METER LED	—	CRAFTECH	1
25	SWITCH BOARD	—	—	1
26	PNEUMATIC CYLINDER	CDQ2A40-50Z	SMC	1
26	FRL	AC20A-02G1-D	SMC	1
27	SOLENOID VALVE	SY7220-5DZ-02	SMC	1
28	FLOW CONTROL	AS2201F-02-08A	SMC	1
29	LOTO SWITCH	VHS20A-02-D	SMC	1

PROWESS
AUTOTECH PVT. LTD.
OUTSOURCE YOUR PAIN

WARRANTY & SERVICE INFORMATION

PROJECT NO.	
DOC. NO.	
REV. NO.	00
DATE	

DETAILS

MACHINE	MANUFACTURER	CUSTOMER
MCU SUB ASSEMBLY STATION	PROWESS AUTOTECH PVT LTD.	TATA MOTORS LTD.

1 WARRANTY



- Warranty covers manufacturing defects only.
- Damage due to improper installation, misuse, unauthorized modification, or poor maintenance is excluded.
- Use only genuine spare parts.
- Warranty starts from the date of commissioning unless otherwise agreed.

2 SERVICE SUPPORT



Technical assistance, commissioning support, troubleshooting and spare parts are available from Prowess Autotech.

3 CONTACT DETAILS



Company: PROWESS AUTOTECH PVT LTD.

Address: Ground Floor, Plot No. 57, Sector IDC,
SIDCUL Industrial Area Road, IIE Pantnagar,
Rudrapur, Udham Singh Nagar, Uttarakhand - 263153

Phone: +91 9917474110

Email: prowessautotech@gmail.com

4 SERVICE REQUEST INFORMATION



DETAILS	INFORMATION
Machine No.	
Project No.	
Installation Date	
Issue Description	
Contact Person	
Remarks	

PROWESS
AUTOTECH
OUTSOURCE YOUR PAIN

DOCUMENT REVISION HISTORY

REVISION	DATE	DESCRIPTION	APPROVED BY
Rev.00	10-10-2026	Initial Release	PROWESS AUTOTECH PVT LTD

NOTES

- This manual contains important information for the safe operation and maintenance of the **MCU SUB ASSEMBLY STATION**
- Read this manual carefully before installation, operation and maintenance.
- Use only genuine spare parts for replacement.
- Follow all safety instructions and maintenance schedules for reliable performance.
- For any service support or technical assistance, contact Prowess Autotech.

SERVICE RECORD

DATE	COMPLAINT / DESCRIPTION	ACTION TAKEN	ENGINEER NAME	SIGNATURE

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THANK YOU FOR CHOOSING PROWESS AUTOTECH

PROWESS AUTOTECH PVT LTD
LOCATION - RUDRAPUR,UTTARAKHAND
PIN CODE : 263153

+91 9917474110

prowessautotech@gmail.com